

Practice Exercise 1- Secure Data Protection Using Encryption in a Java REST API

Objective

Design and implement a secure Java REST API that protects sensitive information using encryption techniques. The application must ensure that confidential data remains protected while being stored and transmitted between client and server.

Background

Organizations often store sensitive personal and financial information that must be protected from unauthorized access. Failure to secure such information can result in identity theft, financial fraud, and privacy violations.

In this assignment, students will implement encryption to protect the following sensitive information:

Sensitive Fields

1. Passport Number

- Used for international travel identification.
- Must be protected from unauthorized access and misuse.

2. Bank Account Number

- Used for financial transactions and banking operations.

- Exposure can lead to financial fraud and unauthorized transactions.

3. Driving License Number

- Used as an official identity document.
- Leakage may result in identity theft and misuse of personal information.

To protect this information, encryption must be applied before storing or transmitting the data.

Assignment Requirements

Task 1: Create a Spring Boot Project

Create a Spring Boot REST application using Maven.

Required Dependency:

- Spring Web

Task 2: Design a Data Model

Create a model named **Citizen**, **Customer**, or **User** containing the following fields:

Field Name	Sensitive
Full Name	No
Passport Number	Yes
Bank Account Number	Yes

Driving License Number	Yes
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Task 3: Implement Encryption Functionality

Develop a utility component that performs:

- Encryption of sensitive information
- Decryption of sensitive information

Requirements:

- Use a standard encryption algorithm.
 - Encrypt Passport Number, Bank Account Number, and Driving License Number.
 - Ensure encrypted values are not readable in plain text.
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Task 4: Develop REST APIs

Create the following APIs:

API 1 – Save User Information

Purpose:

- Accept user details.
- Encrypt all sensitive fields.
- Return encrypted information.

API 2 – Retrieve User Information

Purpose:

- Accept encrypted user information.
 - Decrypt all sensitive fields.
 - Return original information.
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Task 5: Validate Encryption

Verify that:

- Passport Number is encrypted before storage.
 - Bank Account Number is encrypted before storage.
 - Driving License Number is encrypted before storage.
 - Original values cannot be viewed directly after encryption.
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Task 6: Test Using Postman

Perform the following tests:

Test Case 1

Submit original user information and verify that all sensitive fields are encrypted.

Test Case 2

Submit encrypted information and verify that all sensitive fields are correctly decrypted.

Capture screenshots of:

- Request payload

- Encrypted response
- Decrypted response